Gradients with a numerical check

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One readable gradient-check record.

Trace labelled perturbations,repair denominator/order errors and state the numerical evidence limit.

Keep one habit: W1–W3/W5 → selected new gradient check → project coordinate ledger → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Slopes and local change. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Parameters and architecture. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Activation functions. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Use one example with **input** x = 2 and target t = 3. The model predicts y = wx + b, and loss L(w, b) = (2w + b - 3) squared. At w = 1, b = 0, **prediction** is 2, residual is -1 and loss is 1.

  There are two parameters in order [w, b].

- To estimate the partial slope for w, change w only and keep b = 0. To estimate the partial slope for b, change b only and keep w = 1.

  Moving both at once measures change along a combined direction; it does not isolate either **partial derivative**.

- The **gradient** is [-4, -2] in order [w, b]. Both entries refer to the same original parameter vector, not to a partially updated model.

Slopes and local change

> **My remembered idea, example and correction**

Parameters and architecture

> **My remembered idea, example and correction**

Activation functions

> **My remembered idea, example and correction**

Lesson reference for workbook practice

Perturbation from [1, 0]	Loss at minus step	Loss at plus step	Symmetric slope
w only with step 0.1	1.44	0.64	-4
b only with step 0.1	1.21	0.81	-2

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main x2, target 3, y2w+b, L=(2w+b−3)², parameterorder[w, b]. At[1, 0]:y2, e−1, L1, formula[4e, 2e]=[−4, −2]. w-onlyh.1 losses1.44/.64 over.2→−4; b-only 1.21/.81→−2. At[0, 1]e−2, L4, gradient[−8, −4]; at[1, 1]e0, L0, gradient[0, 0]. Combinedtprobe[1+t, t]losses 1.69/.49 over.2→−6, not a partial. Checker spacing.001, tolerance 1e−6, finite numeric/positive-spacing assumptions.

  ReLU centred difference.5at0 doesnotestablish differentiability. Swapping entries mislabels coordinates. Independent x3, target 2 uses different input sensitivity.

- For the lesson model at w = 1 and b = 1, calculate **prediction**, residual, loss and **gradient**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main x2, target 3, y2w+b, L=(2w+b−3)², parameterorder[w, b]. At[1, 0]:y2, e−1, L1, formula[4e, 2e]=[−4, −2]. w-onlyh.1 losses1.44/.64 over.2→−4; b-only 1.21/.81→−2. At[0, 1]e−2, L4, gradient[−8, −4]; at[1, 1]e0, L0, gradient[0, 0]. Combinedtprobe[1+t, t]losses 1.69/.49 over.2→−6, not a partial. Checker spacing.001, tolerance 1e−6, finite numeric/positive-spacing assumptions.

  ReLU centred difference.5at0 doesnotestablish differentiability. Swapping entries mislabels coordinates. Independent x3, target 2 uses different input sensitivity.

- A w probe has losses 1.44 and 0.64 at 0.9 and 1.1 with b fixed. A learner divides their difference by 0.1. Correct the result.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main x2, target 3, y2w+b, L=(2w+b−3)², parameterorder[w, b]. At[1, 0]:y2, e−1, L1, formula[4e, 2e]=[−4, −2]. w-onlyh.1 losses1.44/.64 over.2→−4; b-only 1.21/.81→−2. At[0, 1]e−2, L4, gradient[−8, −4]; at[1, 1]e0, L0, gradient[0, 0]. Combinedtprobe[1+t, t]losses 1.69/.49 over.2→−6, not a partial. Checker spacing.001, tolerance 1e−6, finite numeric/positive-spacing assumptions.

  ReLU centred difference.5at0 doesnotestablish differentiability. Swapping entries mislabels coordinates. Independent x3, target 2 uses different input sensitivity.

- Starting at [w, b] = [1, 0], both coordinates move by plus or minus 0.1 together. Compute the symmetric slope and explain why it is not either **partial derivative**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main x2, target 3, y2w+b, L=(2w+b−3)², parameterorder[w, b]. At[1, 0]:y2, e−1, L1, formula[4e, 2e]=[−4, −2]. w-onlyh.1 losses1.44/.64 over.2→−4; b-only 1.21/.81→−2. At[0, 1]e−2, L4, gradient[−8, −4]; at[1, 1]e0, L0, gradient[0, 0]. Combinedtprobe[1+t, t]losses 1.69/.49 over.2→−6, not a partial. Checker spacing.001, tolerance 1e−6, finite numeric/positive-spacing assumptions.

  ReLU centred difference.5at0 doesnotestablish differentiability. Swapping entries mislabels coordinates. Independent x3, target 2 uses different input sensitivity.

- Compute the symmetric difference of ReLU at zero with step 0.1. Explain why this does not prove a derivative exists there.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use x = 3, target 2 and y = wx + b at w = 0, b = 1. With step 0.1, calculate both numerical partials. Compare with [2 times residual times x, 2 times residual].

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Check a new example independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled **gradient**s with a numerical check **evidence** sheet to Image-pattern Network Investigation. Show how you can estimate one **partial derivative** while other parameters stay fixed.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Catch a denominator error for **Gradient**s with a numerical check. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Gradient**s with a numerical check. Proposed explanation: "The **gradient** is one scalar". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use x = 3, target 2 and y = wx + b at w = 0, b = 1. With step 0.1, calculate both numerical partials. Compare with [2 times residual times x, 2 times residual].

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

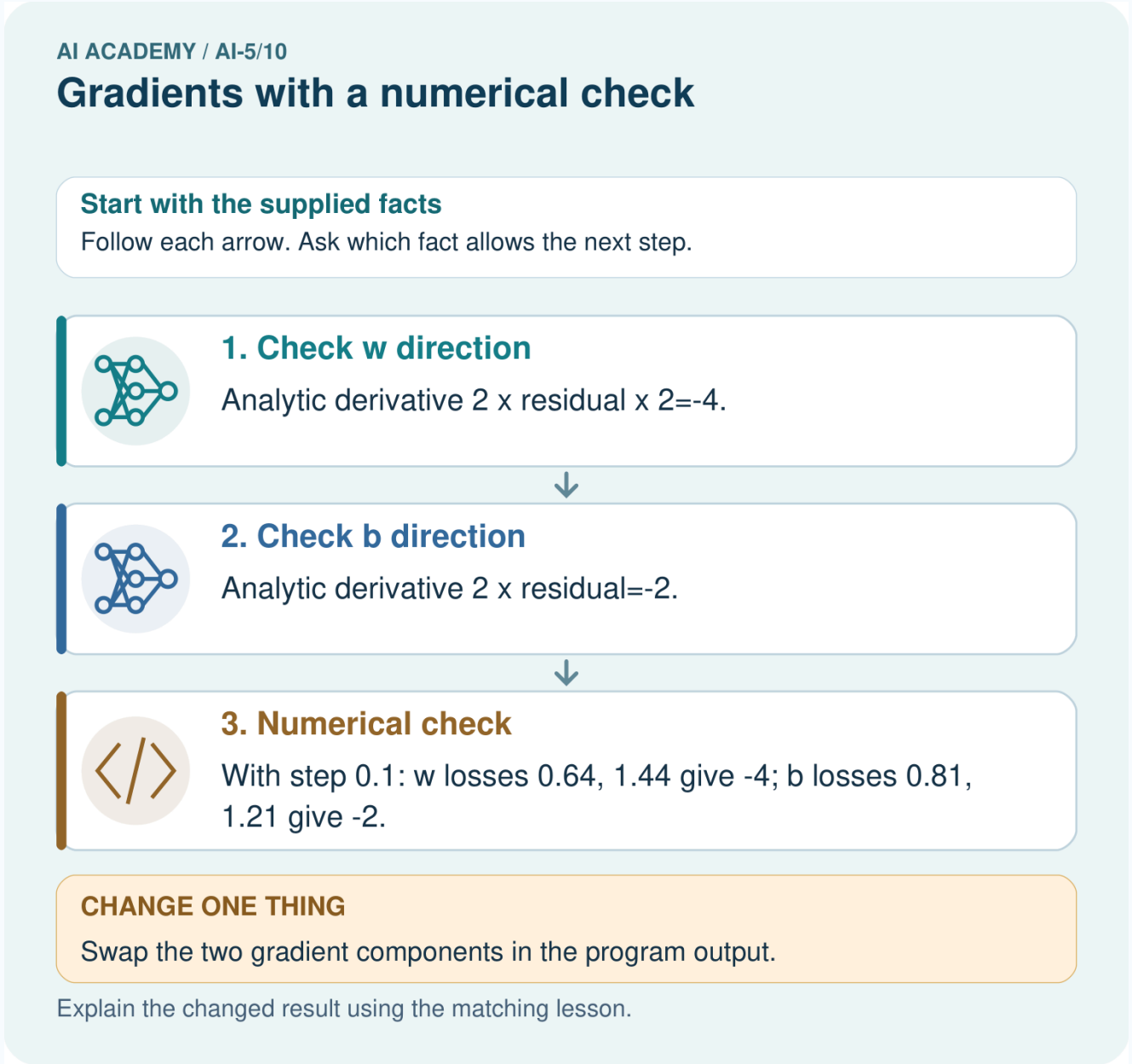
> Keep this task's inputs,rule and response together.

Complete task and evidence

- Check each coordinate of a **gradient** while holding the other parameter fixed.
- Model y=2w+b, target 3. At [w, b]=[1, 0], y = 2, residual y-target=-1; loss is residual².
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Why should the numerical checks perturb one coordinate at a time? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Check each coordinate of a **gradient** while holding the other parameter fixed.
- Which labelled step matches this detail? With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2.
- 1. Numerical check
- 2. Check b direction

- 3. Check w direction

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Numerical check With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2. A **gradient** lists the slope for each parameter, in a declared order.

Changing w by a small amount changes the **prediction** twice as much as changing b by the same amount. That explains the different derivative magnitudes.

Perturb one coordinate while holding the other fixed, then compare the numerical slope with its analytic counterpart.

Changed-case question and optional hint

- Changed-case practice: Swap the two **gradient** components in the program **output**. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. The coordinate-wise numerical check exposes the mismatch.

- 3. We can decide the result without examining the changed **input** or the procedure.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The coordinate-wise numerical check exposes the mismatch. The w perturbation changes y twice as strongly as the same b perturbation.

Repair and retain the record

Review Slopes and local change (ai-5/9). Goal: Calculate an average rate of change over a stated interval. Short practice: Which calculation disproves “any increase helps”?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Parameters and architecture (ai-5/7). Goal: Distinguish architectural choices from parameter values. Short practice: How many new parameters did that one hidden unit add?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Activation functions (ai-5/4). Goal: Trace identity ReLU and sigmoid on a numerical score. Short practice: Which original **input**s cannot be distinguished by this network’s final **output**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |
> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Gradients with a numerical check: the case

AI ACADEMY / AI-5/10 · 01 · LOOK AT THE FACTS

Gradients with a numerical check: the case

The situation

Check each coordinate of a gradient while holding the other parameter fixed.

Supplied evidence

Model y=2w+b, target 3. At [w, b]=[1, 0], y = 2, residual y-target=-1; loss is residual².
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain gradients with a numerical check using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

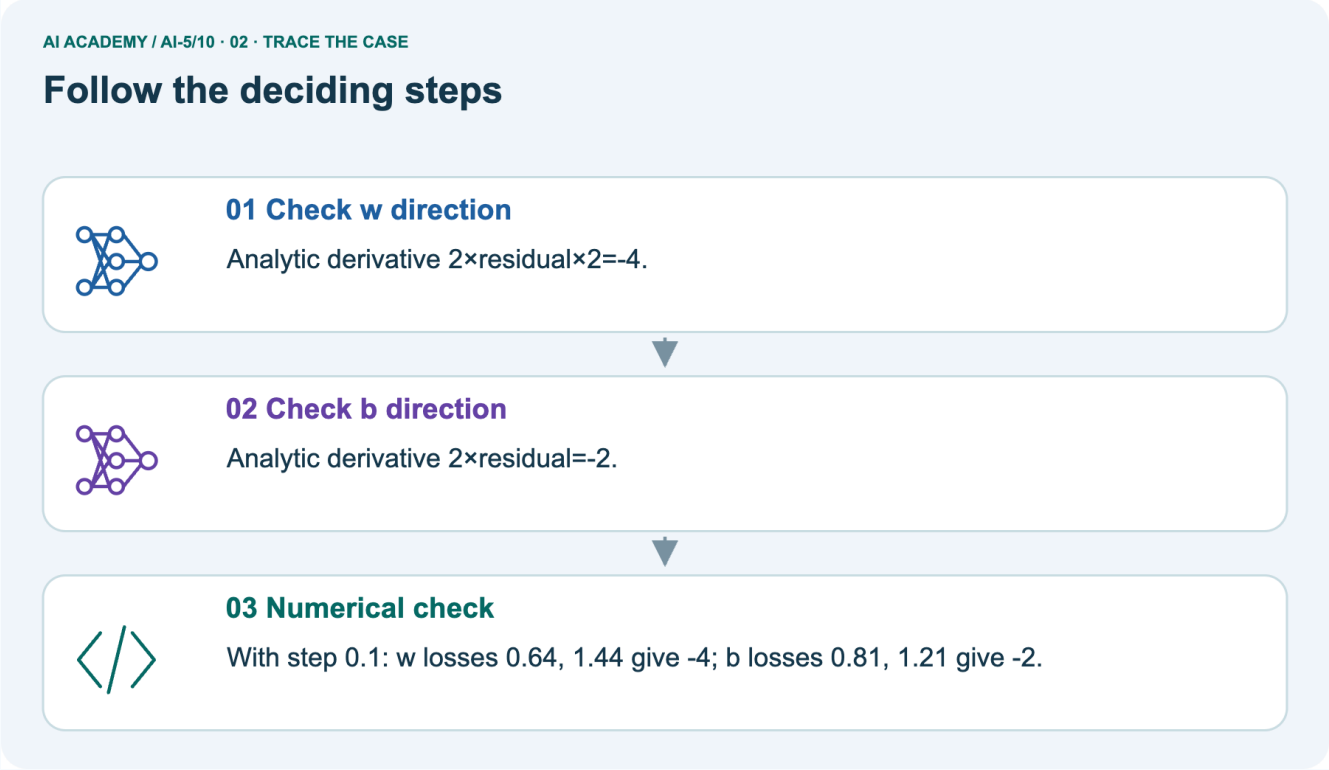
Check each coordinate of a **gradient** while holding the other parameter fixed. Supplied facts: Model y=2w+b, target 3. At [w, b]=[1, 0], y = 2, residual y-target=-1; loss is residual².

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Check w direction: Analytic derivative 2×residual×2=-4. → Check b direction: Analytic derivative 2×residual=-2. → Numerical check: With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Two separate probes at one parameter vector

AI ACADEMY / AI-5/10 · 03 · INSPECT THE LESSON MODEL

Two separate probes at one parameter vector

Parameter being probed	Held fixed	Gradient entry
w near 1	b is 0 and x is 2	First entry -4
b near 0	w is 1 and x is 2	Second entry -2
Assemble in order [w, b]	Same original model	Gradient [-4, -2]

The first two rows are independent local probes, not successive parameter updates. The final row records both results in the declared order.

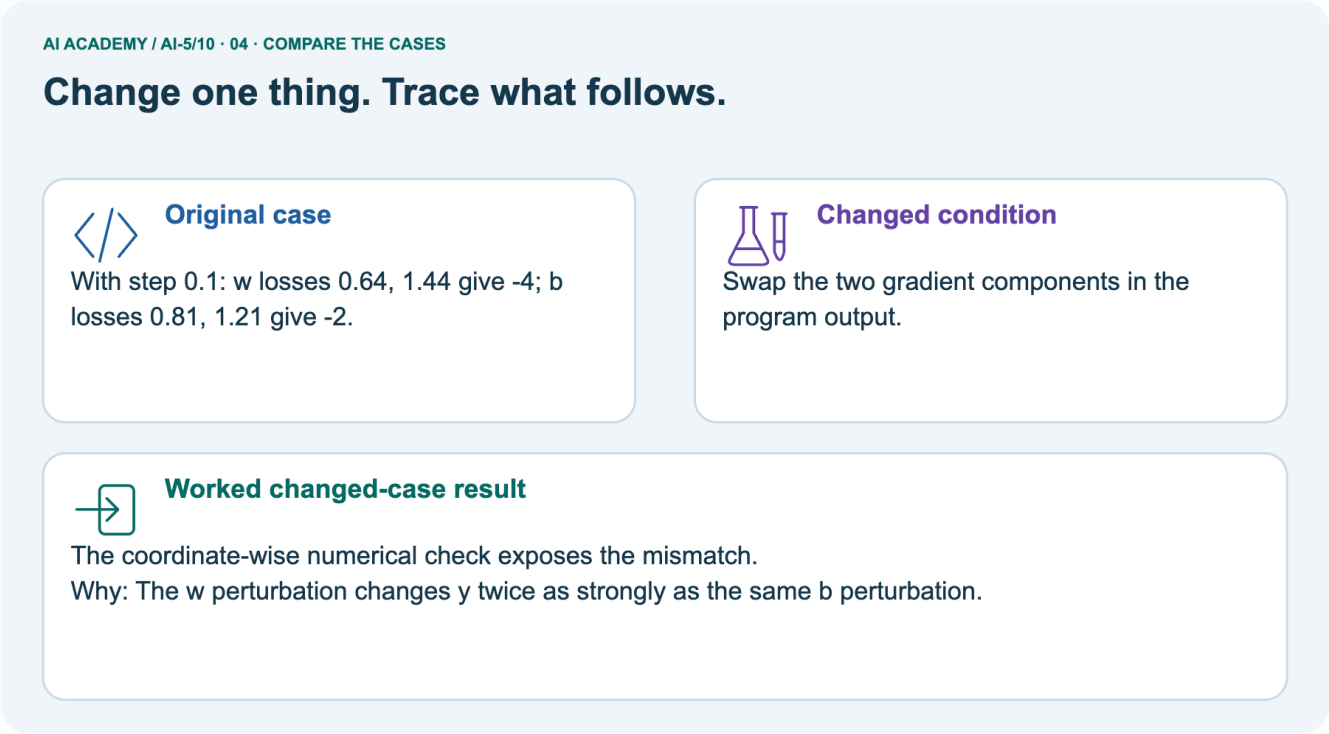
Follow the labelled picture with the matching explanation. Open the vector diagram

Two separate probes at one parameter vector. w near 1 | b is 0 and x is 2 | First entry -4. b near 0 | w is 1 and x is 2 | Second entry -2. Assemble in order [w, b] | Same original model | **Gradient** [-4, -2]

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

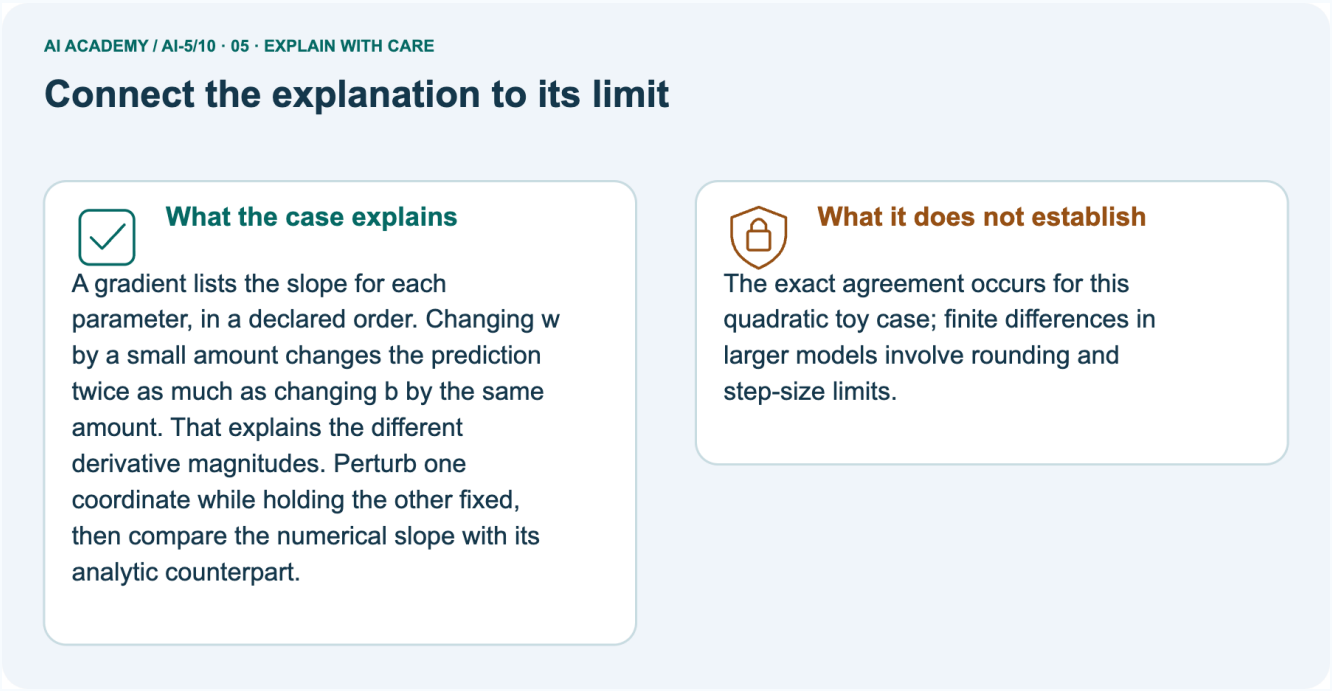
Original case: With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2. Changed condition: Swap the two **gradient** components in the program **output**.

Expected result: The coordinate-wise numerical check exposes the mismatch. Reason: The w perturbation changes y twice as strongly as the same b perturbation.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **gradient** lists the slope for each parameter, in a declared order. Changing w by a small amount changes the **prediction** twice as much as changing b by the same amount. That explains the different derivative magnitudes.

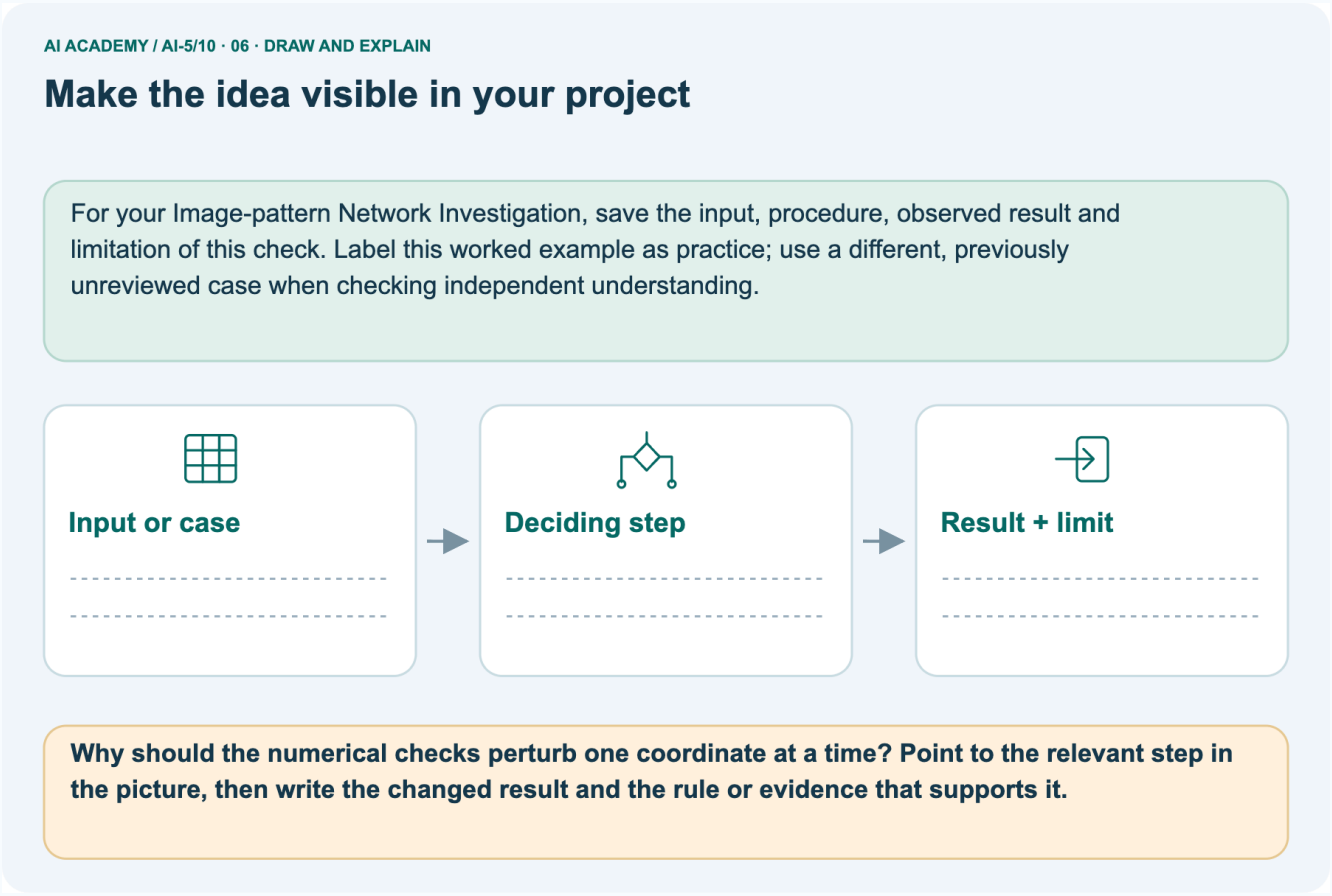
Perturb one coordinate while holding the other fixed, then compare the numerical slope with its analytic counterpart.

Limit: The exact agreement occurs for this quadratic toy case; **finite difference**s in larger models involve rounding and step-size limits.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should the numerical checks perturb one coordinate at a time? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

01. Gradients with a numerical check: the case

02. Follow the deciding steps

03. Two separate probes at one parameter vector

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project
</details>

<details>
<summary>Optional interactive practice and completed work</summary>

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.
</details>